

Keerthirajan Murugesan

Graduate AI Engineer

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Executive Summary

Future-focused AI engineer with an MSc in Artificial Intelligence & Data Science and founder of AstralPrism, a software studio delivering production-grade web platforms, DeFi tools, and enterprise prototypes. Turns messy, real-world data into deployable NLP and ML systems, not just academic prototypes. Built an AI-powered clinical text de-identification pipeline using SpaCy NER and a deep-learning crime forecasting model for London, covering everything from data cleaning and feature engineering to evaluation and model monitoring. Combines production-minded engineering with proven leadership as a retail store manager, bringing rare end-to-end ownership: from understanding users and business constraints to shipping reliable, privacy-aware AI solutions.

Skills

Programming: Python (Advanced, OOP, REST APIs), JavaScript, PHP, SQL, Java, C, HTML/CSS

ML/AI & NLP: spaCy (NER), NLTK, Transformers, TensorFlow, scikit-learn, LLMs (fine-tuning, prompt engineering), Model Evaluation, MLOps

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering, Data Cleaning, ETL

Web & Frontend: React, Next.js, Flask, HTML/CSS, Responsive Design, UI/UX Prototyping

Web3 & DeFi: ethers.js, DeFiLlama API, Ethereum RPC, Flash Loan Mechanics, On-chain Analytics

Trading & Finance: Forex Algorithmic Trading (MT5, MQL5), Crypto Trading, Options Trading, Hedging Strategies (Straddle, Strangle, Iron Condor), Flash Loan Arbitrage, Triangular Arbitrage

Backend & Databases: REST APIs, MySQL, PostgreSQL, SQL/NoSQL, Session Management

Cloud & DevOps: AWS (EC2, S3), Docker, Git, GitHub, CI/CD, Vercel, Jupyter

Personal Skills: Problem-Solving, Analytical Thinking, End-to-End Ownership, Client Communication, Adaptability.

Education

University of Hull, UK

Master of Science: Artificial Intelligence & Data Science

2024 – 2025

KSR Institute for Engineering and Technology (Anna University), India

Bachelor of Engineering: Computer Science & Engineering (Distinction) CGPA: 8.69 / 10

2019 – 2023

Projects

AI-Powered Clinical Text De-identification

MSc Research Project

- Built a spaCy NER pipeline to automatically remove protected health information from clinical text.
- Performed data cleaning, manual annotation, and iterative model training on real-world healthcare data.

Predictive Analysis of Anti-Social Behaviour in Greater London

- Developed ML/DL models to predict crime hotspots across 32 London boroughs.
- Engineered temporal, demographic, and POI-based features to forecast weekly crime trends.

Planet Nutrophilus – Full-Scale E-Commerce Web Platform [Link](#)

- Designed and developed the complete production website for a nutrition startup, end-to-end.
- Built a responsive, scalable platform supporting product catalogue, branding, and customer journey.
- Stack: Next.js, React, HTML/CSS (E-Commerce, Branding, Full-Stack).

Rathna UK Ltd – Enterprise Distribution Prototype [link](#)

- Designed a high-fidelity enterprise prototype for a wholesale distribution management system.
- Streamlined complex B2B wholesale operations into a modern, intuitive interface.
- Stack: Next.js, React (B2B, Prototyping).

Quantum Flow – Algorithmic Trading UI [link](#)

- static dashboard into a professional full-stack Next.js platform featuring real-time market feeds and interactive trading tools like Backtest and Monte Carlo engines.
- Focused on performance and clarity for real-time analytics dashboards.
- Stack: React, analytics libraries (Trading, Analytics).

Full-Stack Web Login System

- Designed a secure authentication system using Python, PHP, MySQL, HTML, and CSS.
- Implemented login/registration workflows with session and database management.

Information Hiding in Images (Steganography Project)

- Designed algorithms to perform reliable message encoding and decoding while minimising perceptual distortion.
- Gained hands-on experience with image processing, binary data handling, and algorithm optimisation.

MyFunValentine – Interactive Web Experience [link](#)

- MyFunValentine is a small, fun full-stack project I built using HTML, CSS, JavaScript, and Flask.
- It's an interactive web app designed to be playful and engaging, and it helped me practice frontend creativity while connecting it to simple backend logic.

DeFi Flash Loan Educational Dashboard [Link](#)

- Created interactive tool demonstrating Aave flash loans, arbitrage realities (gas/slippage/MEV), liquidations.
- Used Next.js, ethers.js, DeFiLlama API, and real-time Ethereum RPC integration.

Work Experience

Founder & Lead Engineer

[astralprism](#)

AstralPrism – Hull, UK

2026 – Present

- Founded and operate a software studio building production-grade web platforms, DeFi tools, and enterprise prototypes for real clients.
- Manage the full delivery lifecycle: client scoping, UI/UX design, development, and deployment.
- Delivered projects including a nutrition e-commerce platform, a DeFi analytics terminal, a B2B enterprise prototype, and an algorithmic trading UI.

Sales Manager

Morrisons Daily – Hull, UK

Jan 2025 – Dec 2025

- Led a small retail team in daily operations, performance oversight, ensuring smooth store functionality.
- Managed stock levels, addressed customer issues, and handled documentation, which streamlined operations.

Shop Assistant

Morrisons Daily – Hull, UK

Jan 2024 – Dec 2024

- Maintained stock rotation and ensured compliance with labelling standards to optimize inventory management.
- Delivered high-quality customer service and assisted with till operations, enhancing customer satisfaction.

Leadership & Activities

- President, E-CAP Club (KSRIET): Organised technical workshops and coding events.
- Demonstrated leadership through retail team management and staff mentoring.